ML-Driven Inverse Design of Microring Resonators for Co-Packaged Optics / High-Bandwidth Optical Interconnects

Md Shidul Islam Shihab*[1]

[1]Department of Electronics & Communication Engineering, Khulna University of Engineering & Technology (KUET), Khulna, Bangladesh

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Abstract

Deep-learning inverse design of photonic components rests on an unresolved disagreement. Liu *et al.* [3] showed that a naive, non-tandem inverse network can fail to converge when the design space is redundant; Chen and Jiang [7] and Adibnia *et al.* [9] reported naive inverse networks converging acceptably in their own design spaces. This work shows that the two reports are consistent with each other for a four-parameter, bus-coupled add-drop silicon microring resonator, by measuring both sides on one dataset with one set of metrics. We build a dual-fidelity dataset of 1,150 labelled geometries over ring radius, waveguide width, coupling gap and coupling length, with the coupling coefficient obtained from 200 three-dimensional FDTD simulations of the coupler alone and the ring response computed analytically, and we validate the construction against a published, experimentally characterised device, against 68 further independent FDTD coupler runs, and against nine full 3-D simulations of the complete ring (three geometries at three mesh or sampling settings, all at $R = 5.2\,\mu\text{m}$).

The naive inverse network *fails* the geometry-space criterion ($R^2 \geq 0.80$ on only one of four parameters) and *passes* the response-space criterion (closed-loop re-simulated error 3.669% on twenty test designs, all physically realisable, and 3.989% on a hundred, of which 96 were realisable). Both prior reports are therefore consistent with these results, and were measuring different things. We explain why by counting constraints: of four target quantities, resonance wavelength carries almost no geometric information ($R^2 = 0.0078 \pm 0.0011$ across five seeds), free spectral range fixes the round-trip length, and loaded Q and insertion loss both act through a single coupling phase θ — two effective constraints for four unknowns, leaving a two-dimensional degenerate family.

Three further results follow. First, geometry-space error does not predict response-space error: across four models and three architecture families, closed-loop error is proportional to θ error to within a factor 1.20, while the same ratio formed with RMS-RGE varies by a factor 4.00. Second, physical realisability is architecture-dependent and is invisible to both metrics — Random Forest emits buildable geometries in 100% of cases, provably so by convexity, against 95.4% for a plain network and 33.5% for a tandem network (11.6–83.8% over five initialisations, and 100% once its outputs are bounded). Third, a reproducible information-bound estimate computed from a 10^6-geometry response bank gives RMS-RGE = 14.626%. This point estimate lies below the 15% objective threshold, so we find no evidence that the threshold is unreachable; the margin (0.37 pp) is smaller than the sampling uncertainty expected for 173 test targets, and no interval has yet

*mdshidulislamshihab@gmail.com

been computed. Translating the designs into two co-packaged-optics scenarios, no method meets any scenario's full specification within the uncertainty of the analytic ring model (one marginal case is undecided); eleven of twelve failures involve resonance placement, whose closed-loop error of 4.32 nm RMS exceeds a ±2.0 nm channel tolerance by a factor of two and a ±0.4 nm tolerance by eleven, and is no better than always predicting the dataset's mean resonance wavelength (2.66 nm RMS on the same twenty targets, 3.11 nm over the dataset). Absolute resonance placement in this design space is beyond inverse design and requires post-fabrication trimming.

1 Introduction

Inverse design asks for a geometry given a desired optical response. Where a forward simulation is expensive, a natural approach is to train a surrogate on simulated data and then invert it, and a substantial literature now does exactly this for nanophotonic components [1,2]. Within that literature sits a specific, unresolved disagreement about whether the simplest approach works.

Liu *et al.* [3] argued that training a network to map response directly to geometry fails when several geometries produce the same response: the network is asked to fit a one-to-many relation and converges to an average that is itself not a solution. Their remedy, the tandem network, trains the inverse network through a frozen forward model so that the loss is measured in response space, where the relation is single-valued. This argument is widely accepted and widely cited.

Against it, two later studies report the naive approach working. Chen and Jiang [7] inverse-designed a four-ring channel-dropping filter with two independently trained networks — no tandem — and reported good convergence, validating several designs by re-simulation. Adibnia *et al.* [9] likewise reported acceptable convergence for a direct inverse network. Neither is a refutation of [3], and [3] is not a refutation of them, because the three studies work in different design spaces and, critically, judge success by different criteria: [3] reports geometric accuracy, while [7] and [9] report how closely a re-simulated design reproduces the requested response.

No prior study we are aware of tests this for a bus-coupled add-drop microring resonator. Pal *et al.* [6] apply classical tree-based machine learning to microring inverse design on 462 finite-element samples, and two recent preprints apply machine learning to microrings with different aims. Simsek *et al.* [15] use neural networks to recover the width and height of a silicon-nitride ring from its dispersion data, a characterisation problem with no coupling section. Hasani *et al.* [16] combine a generative network with gradient boosting on FDTD data for a silicon ring parameterised by radius, gap and width, and report RMSE together with one FDTD-checked high-Q design. The design space of Pal *et al.* [6] contains no bus-coupling parameter at all and so cannot predict insertion loss or any other coupling-dependent quantity — the specifications that decide whether a filter is usable in a co-packaged-optics (CPO) link. Torabi *et al.* [12], the most recent and most dimensionally comparable study, defines the RMS-RGE metric adopted here but addresses a polymer-scale biosensor ring and includes no classical-machine-learning baseline; it is also an unrefereed preprint and is cited as such throughout. Valencia-Garzón *et al.* [10] release a large all-pass dataset without modelling it.

This paper contributes the following.

1. **A test of the disagreement**, measured on one dataset with one protocol: the naive inverse network fails in geometry space and succeeds in response space, so the two reports are consistent. We give a closed-form reason, by counting how many independent constraints four scalar targets actually impose.
2. **Evidence that geometry-space error is the wrong metric**, and a physically motivated replacement that predicts response-space error across architectures.

Table 1: Design space and target quantities.

Symbol	Parameter	Range	Note
R	ring radius	5 µm to 15 µm	spans FSR ≈ 5.7 nm to 18.2 nm
w	waveguide width	400 nm to 500 nm	single-mode strip on 220 nm SOI
g	coupling gap	150 nm to 350 nm	near-critical to weakly coupled
L_c	coupling length	0 µm to 3 µm	$L_c = 0$ is a valid design point
λ_{res}	resonance wavelength	nm	nearest to 1550 nm
FSR	free spectral range	nm	adjacent-resonance spacing
Q_L	loaded quality factor	—	λ_{res}/FWHM
IL	insertion loss	dB	drop port, on resonance

3. **A measurement of physical realisability as a first-class result**, with a proof that one model family cannot violate the design box.
4. **A reproducible information-bound estimate** for this design space, intended to separate model limitations from information limits; at the bank size reached it does not yet carry an interval.
5. **A controlled four-way comparison** of Random Forest, XGBoost, a plain deep network and a tandem network on an identical split with identical metrics, including a sample-efficiency characterisation — the comparison absent from [4,6,7,9,12].

2 Device, design space and methods

2.1 Device and parameterisation

The device is a two-bus, four-port add-drop racetrack microring resonator on silicon-on-insulator: silicon core of fixed height 220 nm, SiO_2 cladding, fundamental TE mode only, scanned over 1500 nm to 1600 nm with a nominal target of 1550 nm. Both couplers share a gap and coupling length, keeping the design space at exactly four dimensions. Dispersive material models are used throughout; no constant index is assumed. A racetrack rather than a circular ring makes the coupling length a genuine continuous parameter, which is the degree of freedom absent from [6], [10] and [12].

Table 1 lists the four free parameters and the four targets. The round-trip length is $L = 2\pi R + 2L_c$.

2.2 Dual-fidelity dataset construction

A full three-dimensional FDTD simulation of the complete ring was measured at roughly 35 h per sample, which puts a 1,000-sample dataset beyond any realistic budget. A two-dimensional simulation runs in 149 min but returns a loaded Q too high by a factor of 11.3, because it cannot represent vertical confinement.

The construction used instead simulates in three dimensions only the part of the device that genuinely requires it. A single pass of the directional coupler is simulated in 3-D FDTD to obtain the power coupling coefficient κ^2; the ring response is then assembled analytically from coupled-mode theory [13,14], with the effective and group indices taken from a 3-D mode sweep and the propagation losses from a bend-loss sweep. The total cost is approximately 16 h.

A five-parameter model is fitted to the coupler simulations,

$$\kappa^2 = \sin^2\theta, \qquad \theta = \kappa_0(w)\, e^{-\gamma(w) g} \left(L_c + B\sqrt{2\pi R/\gamma} \right), \tag{1}$$

Table 2: Validation of the dual-fidelity construction. The out-of-sample κ^2 tests are the coupler simulations performed during closed-loop re-verification (Section 3.3); they were not available to the fit.

Test	Quantity	Result	Criterion
Published device [11]	FSR, 440 nm×220 nm, $R = 20\,\mu$m	4.4008 nm vs 4.400 nm measured	$\leq 10\%$ (O1)
	implied n_g	4.3443 vs 4.3451	—
κ^2 model	8-fold cross-validated MAPE	3.36%, $R^2 = 0.9976$	—
κ^2 model	68 out-of-sample FDTD runs	MAPE 1.23–5.16%, all within 10%	—
	systematic bias	-0.36 to $+3.33\%$	—
Energy conservation	$\kappa^2 + t^2$	1.0000 on every run	—

in which the gap and the coupling length enter *only* through the single phase θ. Equation (1) is the origin of the degeneracy analysed in Section 3.2, and θ is the quantity used as a metric in Section 3.3. The fitted $B = 1.0211$ is close to its first-principles value of unity.

The coupler set was built in two stages by augmented Latin hypercube sampling: 48 points, then a further 152, for 200 in total. The extension widened the κ^2 range from 0.0021–0.3138 to 0.0019–0.4599 and moved B from 1.0425 toward unity, with residuals flat across two and a half decades and no evidence of $\sin^2\theta$ saturation.

The modelling dataset comprises 1,150 labelled geometries — 1,000 from a primary Latin hypercube design plus a 150-point supplementary batch covering a sparse corner of the space — split 805/172/173 into training, validation and test sets with a fixed seed. The test set was used once.

2.3 Validation of the construction

Table 2 summarises three independent checks. The predicted free spectral range of the fabricated device reported by Tang *et al.* [11] agrees with the measured value to within the precision of the published figure (quoted as 4.4 nm, i.e. $\pm 1.1\%$), well inside the 10% requirement. Two caveats must accompany this validation. Only FSR is tested, and FSR depends on group index and round-trip length alone, so the test exercises the mode solver and the length convention and says nothing about the coupling model, the bend-loss model, Q_L or IL. Further, the propagation loss used in the analytic model (1.3 dB/cm) is itself taken from [11]; it does not enter the FSR prediction, so this particular comparison is unaffected, but [11] is not an independent source for the loss model. We also note that the loaded $Q \approx 3 \times 10^5$ appearing in [11] is a stated assumption in a scalability projection, not a measurement, and is therefore not used here.

The κ^2 model was further tested on 68 coupler geometries that arose during closed-loop re-verification and were never seen by the fit. Every one fell within 10%, with a consistent positive bias of a few per cent that should be reported as a known systematic.

2.4 Full 3-D validation of the ring model

The checks above validate the *coupler* and the *mode solver*. The coupled-mode assembly that turns κ^2, n_g and loss into Q_L and IL is not tested by either, and closed-loop re-verification does not test it because it re-simulates the coupler and then recomputes the ring with the same analytic model that produced the labels. Since Q_L and IL carry the degeneracy argument of Section 3.2, we simulated the *complete ring* in 3-D FDTD.

Nine runs were performed: three geometries at $R = 5.20\,\mu$m, $w = 450$ nm, spanning the coupling range, each at the production 32 nm mesh, at a coarse 40 nm mesh, and at 40 nm with three times

Table 3: Full 3-D ring simulation against the analytic model. Richardson extrapolation assumes second-order convergence, which two mesh sizes cannot verify.

	simulated Q_L				analytic	
Geometry	32 nm	40 nm	6000 pts	$h \to 0$	Q_L	error
$g = 150$ nm, $L_c = 2.00$ µm	1780	1749	1751	**1833**	1855	**−1.2%**
$g = 170$ nm, $L_c = 1.00$ µm	4015	3892	3921	**4182**	4264	**−1.9%**
$g = 180$ nm, $L_c = 0.60$ µm	4603	5229	5253	3447	6351	−45.7%

denser spectral sampling as a control on the extraction. The simulation window was sized from the predicted Q_L so that the linewidth exceeds the transform limit of the DFT window by at least a factor three, and auto-shutoff was disabled so that the window is exactly the one assumed. All nine met the resolution criteria (6.1–52.9 points across the FWHM, 3.14–4.12 times the transform limit).

Three findings follow, and they are not all favourable.

Resonance wavelength and free spectral range behave as the published-device check predicted. λ_{res} agrees to 0.10% across all nine runs. FSR is biased +2.2 to +2.8%, the same sign and magnitude every time. Since FSR $= \lambda^2/(n_g L)$ and $L = 2\pi R + 2L_c$ is exact geometry — the ring is built as two straights of length L_c joined by two 180° bends — the discrepancy must lie in n_g: the product $n_g L$ the model uses differs from the ring's by about 2.5%.

The obvious explanation is that the model takes n_g from a *straight-waveguide* mode sweep while a 5.2 µm bend is not a straight waveguide. *That explanation does not survive measurement.* A 210-point bent-waveguide FDE sweep (six radii from 5 µm to 15 µm, six widths, five wavelengths, 10 nm mesh, PML on all boundaries) gives n_g of the bent guide within 0.08% of the straight guide at every point in the grid, and n_{eff} within 0.05%. The bend contribution is roughly thirty times too small to account for the observed bias.

Stated as a percentage of FSR this comparison is treacherous, because FSR itself varies by about 4% across 1500 nm to 1600 nm; a model FSR evaluated at λ_{res} and a measured spacing whose resonance pair is centred 8 nm away are not the same quantity. We therefore compare group indices at matched wavelengths. Each adjacent resonance pair gives $n_g = \lambda_1\lambda_2/[L(\lambda_2 - \lambda_1)]$, which we compare with the FDE value interpolated at that pair's midpoint. Over ten pairs spanning the three geometries, **the ring's group index is** 2.8–3.0% **below the mode solver's**: −2.83% (s.d. 0.24%) when the solver's $n_{\text{eff}}(\lambda)$ is fitted with a quadratic and −3.03% (s.d. 0.31%) with a cubic (ddof = 1; the pairs come from three geometries and are not independent). The deficit has a mild wavelength trend (about −2.4% at 1520 nm to −3.2% at 1567 nm). We do not use the solver's reported n_g column as the reference because it is not smooth in wavelength (below); it gives −2.73% with s.d. 0.72%.

Four candidate explanations are excluded by measurement. *Round-trip length:* the straight fraction $2L_c/L$ is 3.5%, 5.8% and 10.9% for the three geometries — a factor of three — and the deficit is identical for all three, whereas a length error or a straight/bend mixing error would scale with it. *Bend effects:* 0.08%, above. *FSR extraction:* an independent re-extraction from the saved spectra agrees with the production analysis to 0.03%, 0.005% and 0.014%. *The n_g polynomial:* 0.26% at these geometries. What remains is either FDTD numerical dispersion at this mesh or a limitation of the coupled-mode assembly itself; refining the mesh from 40 nm to 32 nm moves FSR *away* from the model rather than toward it, which argues against a simple convergence artefact, and distinguishing the two would require a ring simulation at a substantially finer mesh than was affordable here. We note, without claiming it, that the mode-solver n_g agrees with the measured device of [11] (4.344 against 4.345 implied), so an error on the FDTD side is among the candidates.

Two caveats attach to the sweep itself. The FDE solver intermittently returns a degenerate result from an identical call sequence, so each solve is verified and repeated until a guided mode is found. And the solver's reported n_g is not smooth in wavelength: successive steps for the straight 450 nm guide are +0.0085, +0.0878, +0.0031, −0.0250, one of them an order of magnitude larger than its neighbours. Against a finite-difference n_g computed from the solver's own $n_{\text{eff}}(\lambda)$ the reported values scatter with a standard deviation of 0.54%. We therefore do *not* regenerate the dataset labels from these values: the correction they would apply (0.26%) is smaller than the noise they would inject.

The validation itself was mesh-limited, and at low to moderate Q_L the analytic model is correct to within 2%. Simulated Q_L depends on the mesh, so the 32 nm spacing — established as converged for κ^2, a single-pass quantity — is not converged for Q_L, which accumulates loss over hundreds of round trips. For the two lower-Q_L geometries, refinement moves the simulation *toward* the analytic prediction and a second-order extrapolation to zero mesh recovers it to 1.2% and 1.9% (Table 3). A discrepancy of 4–6% at the production mesh is therefore attributed to numerical error for these two geometries; with only two geometries and two mesh sizes this is an inference, not a convergence proof.

At the highest Q_L the discrepancy is real and unexplained. For the third geometry, refinement moves the simulation *away* from the model. The tripled spectral sampling changed Q_L by 0.45%, so this is neither an extraction nor a sampling artifact. One candidate is that the bend-loss model is a function of radius alone and carries no wavelength dependence, while this geometry resonates at 1555 nm against 1542 nm and 1543 nm for the other two; bend radiation at this radius increases steeply toward longer wavelengths, so a wavelength-blind loss model would over-predict Q_L for the reddest resonance, and a finer mesh resolving genuine radiation would lower it. We state this as a hypothesis: testing it requires a wavelength-resolved bend-loss study that we have not performed.

Insertion loss does not validate. Drop-port peak transmission differs from the analytic prediction by −12.5%, +1.5% and −17.9% at the production mesh, with no consistent sign. One of the three is not evidence at all: that geometry's IL is 0.084 dB, i.e. $T = 0.981$, within 2% of unity, where the absolute uncertainty of extracting IL from a simulated spectrum is comparable to the value itself.

What this does and does not affect. Every model compared in Section 3 was trained and scored on identical labels, so a bias shared by all of them cancels from every model-to-model comparison: the assessment of O6, the metric result of Section 3.3, the realisability result and the sample-efficiency comparison are unaffected. What must be qualified is the absolute accuracy of Q_L above $Q_L \approx 4300$, the absolute accuracy of IL anywhere, and the scenario verdicts, which are judged against Q_L specifications. Of the 1,150 labels, 84.7% have Q_L above 4300 and 76.3% above 6351 (the analytic Q_L of the highest-Q_L geometry simulated), so most absolute Q_L and IL labels lie outside the range this validation supports.

2.5 Models and protocol

Four method families are compared, each in the forward and inverse direction, on the identical split with identical metrics: Random Forest, XGBoost, a plain feed-forward network, and a tandem network trained through the frozen forward model. Tree models are evaluated with raw and normalised inputs. The forward network architecture was selected by a 144-configuration grid search over depth, width, learning rate and weight decay.

Two metric conventions require statement. Aggregate geometric error is the root-mean-square

relative geometric error

$$\text{RMS-RGE} = 100\% \times \sqrt{\frac{1}{4} \sum_{p \in \{R,w,g,L_c\}} \left(\frac{p^{\text{pred}} - p^{\text{target}}}{s_p} \right)^2}, \tag{2}$$

with $s_p = p^{\text{target}}$ for R, w and g following [12], and $s_p = 3\,\mu\text{m}$, the range width, for L_c — necessary because $L_c = 0$ is a valid design point at which a target-normalised relative error is undefined. Per-parameter terms and the pure [12]-convention value over $\{R, w, g\}$ are reported alongside every aggregate.

Closed-loop response error is measured by taking a model's predicted geometry for each of twenty representative test targets (a hundred for the naive network, Section 3.2), re-simulating the coupler in 3-D FDTD, reassembling the ring response, and comparing to the response originally requested. The FDTD step is what makes this independent: feeding predictions back through the fitted κ^2 model would merely replay the model that produced the labels.

3 Results

Table 4 summarises the outcome against the seven objectives fixed before any modelling began. Three are met, three are not, and the central one is consistent with both prior reports, in the direction that the methodology document anticipated as publishable. The objectives that are not met are reported as measured, with the reason in each case tested rather than asserted.

3.1 The forward surrogate, and an unlearnable target

The best forward model is the network, at 3.568% mean MAPE across the four targets against 5.323% for the best XGBoost variant and 11.929% for the best Random Forest. Table 5 reports all three required metrics for every target and method.

Objective O2 required $R^2 \geq 0.90$ and MAPE $\leq 10\%$ for at least three of four targets. **Two of four are met, so O2 is not met.** Two points about that verdict deserve care.

First, the verdict is reproducible, and it was measured as such. λ_{res} fails on R^2 by an enormous margin and does so consistently; IL fails on MAPE at $11.331 \pm 1.101\%$ against a 10% bar, with only one seed in five passing. A single training run is not a sound basis for a hard-threshold verdict, and we report the distribution rather than a draw from it.

Second, IL's failure is a property of the metric and the target's scale, not of the model's accuracy. IL is predicted to 0.072 dB RMSE with $R^2 = 0.9966$. But the dataset's IL has a median of 0.524 dB, with 48.3% of rows below 0.5 dB and a minimum of 0.032 dB, so a constant absolute error of 0.07 dB produces 13.7% relative error at the median and over 200% at the extreme. Clearing a 10% MAPE bar would require absolute accuracy better than about 0.052 dB on a decibel quantity already predicted to seven hundredths of a decibel. We report O2 as not met under the criterion as defined. Whether that verdict carries system-level meaning is a separate question, and this work can answer it from its own scenario definitions rather than by appeal to typical practice: Scenario A allows 1.5 dB of insertion loss per ring and Scenario B 1.0 dB, so a 0.072 dB RMSE consumes 4.8% and 7.2% of those budgets respectively. The model is accurate enough for both. A relative-error threshold was the wrong instrument for a target whose values approach zero, and we would specify an absolute one — tied to the link budget, as above — if fixing the objectives again.

The resonance wavelength is not learnable by any method here. All four tree variants return *negative* R^2 — worse than predicting the mean — and the network manages 0.0069. The cause is

Table 4: Objective outcomes. O6 is the central contribution; O1–O5 and O7 exist to support its assessment rather than to stand as independent results of equal weight.

	Outcome	Basis
O1 dataset + validation	met	1,150 geometries ($\geq$ 800 required); FSR agrees with [11] to within the published figure's precision, well inside 10%
O2 forward surrogate	**not met**	2 of 4 targets; IL MAPE 11.331 ± 1.101% against $\leq$ 10% over five seeds. Cause: small denominators on a sub-decibel target, at 0.072 dB RMSE
O3 inverse accuracy	**not met**	RMS-RGE = 15.506% against $\leq$ 15%; closed-loop half met at 3.669% (3.989% over a hundred targets). Information-bound estimate 14.626% ($<$ 15%): no evidence the threshold is unreachable
O4 four-way comparison	met	all four families, identical split, every metric, no omissions
O5 sample efficiency	met	XGBoost forward below 300 samples; network at full size; naive network inverse at both (ordering holds across seeds; at full size the paired test-row interval against RF includes zero)
O6 one-to-many	**consistent**	geometry-space criterion fails (1 of 4 parameters at $R^2 \geq 0.80$), response-space criterion passes (3.669%) — both prior reports consistent with these data
O7 CPO translation	**not met**	0 of 12 candidates, re-simulated (coupler in 3-D FDTD, ring analytic); 11 of 12 failures involve resonance placement, which the models do not predict better than the dataset mean

sensitivity: $\mathrm{d}\lambda_{\mathrm{res}}/\mathrm{d}R \approx 0.15\,\mathrm{nm/nm}$, so placing a resonance to within 1 nm requires the radius to roughly 7 nm, about one part in 1.5×10^4 of the range. The consequence propagates: λ_{res} accounts for almost all of the forward model's residual loss, so the 144-configuration grid search was in effect ranking configurations on noise.

3.2 The one-to-many question

Objective O6 requires the naive inverse network to be judged against *both* criteria used by the two sides of the disagreement: a geometry-space criterion of $R^2 \geq 0.80$, and a response-space criterion of closed-loop re-simulated error.

In geometry space the naive network fails. Its per-parameter R^2 values are 0.9930 for R, 0.2920 for w, 0.6914 for g and 0.2166 for L_c: one parameter of four clears 0.80. Figure 1 shows that this is not specific to the network — every method recovers the radius and no method recovers the other three.

In response space the naive network passes, comfortably. Re-simulating its predicted geometries for twenty representative targets gives a root-mean-square closed-loop response error of 3.669%, against a 20% requirement, with all twenty predictions physically realisable. Per target: λ_{res} 0.203%, FSR 0.980%, Q_L 4.037%, IL 3.594%. Twenty targets is a small sample, so we extended the check for the naive network to a hundred test targets spaced evenly through the test set (twelve of them are among the original twenty). The result holds, with one correction. The root-mean-square closed-loop error over the 96 realisable predictions is 3.989% (bootstrap 95% interval 3.36–4.62%,

Table 5: Forward-direction accuracy on the held-out test set ($n = 173$). Every method appears for every target. The best network is reported as a mean over five training seeds, since the objective threshold is a hard bound and a single run cannot distinguish a crossing from initialisation noise.

	R^2				MAPE (%)			
Method	λ_{res}	FSR	Q_L	IL	λ_{res}	FSR	Q_L	IL
DNN	0.0069	0.9993	0.9980	0.9969	0.158	0.561	3.205	10.346
RF (raw)	−0.0950	0.6892	0.9504	0.9342	0.166	17.288	12.640	25.351
RF (norm)	−0.0688	0.9920	0.9147	0.9072	0.164	2.730	17.340	27.483
XGB (raw)	−0.0837	0.9992	0.9634	0.9494	0.168	0.699	8.811	12.640
XGB (norm)	−0.0795	0.9992	0.9808	0.9489	0.168	0.695	7.701	12.727
DNN over five seeds (42–46), mean ± s.d.:								
λ_{res}	$R^2 = 0.0078 \pm 0.0011$				MAPE $= 0.158 \pm 0.000$			(0/5 seeds pass)
FSR	$R^2 = 0.9993 \pm 0.0001$				MAPE $= 0.556 \pm 0.020$			(5/5 pass)
Q_L	$R^2 = 0.9982 \pm 0.0001$				MAPE $= 3.059 \pm 0.109$			(5/5 pass)
IL	$R^2 = 0.9966 \pm 0.0003$				MAPE $= 11.331 \pm 1.101$			(1/5 pass)

Table 6: Inverse-direction accuracy, all methods on the identical test split. RMS-RGE is Eq. (2); per-parameter terms and the $\{R, w, g\}$-only value are given as the metric convention requires. "Buildable" is the fraction of test predictions lying inside the design box.

	RMS-RGE (%)						
Method	aggregate	R	w	g	L_c	R, w, g only	Buildable
naive DNN	**15.506**	2.841	5.321	14.144	26.930	8.878	95.4%
Tandem	29.461	6.079	7.852	15.296	56.029	10.529	33.5%
RF (raw)	16.039	5.942	5.741	14.378	27.460	9.574	100%
RF (norm)	15.835	3.730	5.926	14.213	27.421	9.148	100%
XGB (raw)	16.451	2.949	5.925	14.847	28.607	9.385	100%
XGB (norm)	16.546	2.986	5.941	14.687	28.900	9.308	100%

$B = 10\,000$), and every target of every realisable design is within 20% of the request, the worst single error being 17.98%. Per target (mean absolute error): λ_{res} 0.212%, FSR 0.879%, Q_L 3.936%, IL 4.452%. The correction is that *not all designs are buildable*: four of the hundred predictions fall just outside the design box (radius 4.87–4.99 µm against a lower limit of 5 µm), so the buildable fraction is 96% (Wilson 95% interval 90.2–98.4%). This agrees with the 95.4% of Table 6 and shows that the "all twenty realisable" result above was a feature of the small sample. The 96 coupler simulations of this run were also not in the κ^2 fit; the fitted model sits above them by 3.32% on average (MAPE 4.10%, maximum 8.72%, all within 10%), consistent with the positive bias of a few per cent noted in Section 2.3. The other models were not extended and are compared at twenty targets in Section 3.3.

This is the outcome in which the two literatures are simultaneously consistent with the data. The reports differ in the criterion they apply; for this device the disagreement is one of criterion, not of measured result.

Why: counting constraints. The structure of the problem explains the split exactly. Of the four targets, λ_{res} carries essentially no geometric information, as Section 3.1 established. FSR fixes the round-trip length $L = 2\pi R + 2L_c$. And Q_L and IL both depend on the coupling only through $\kappa^2 = \sin^2\theta$, so they constrain the single phase θ of Eq. (1) and nothing finer. Four targets

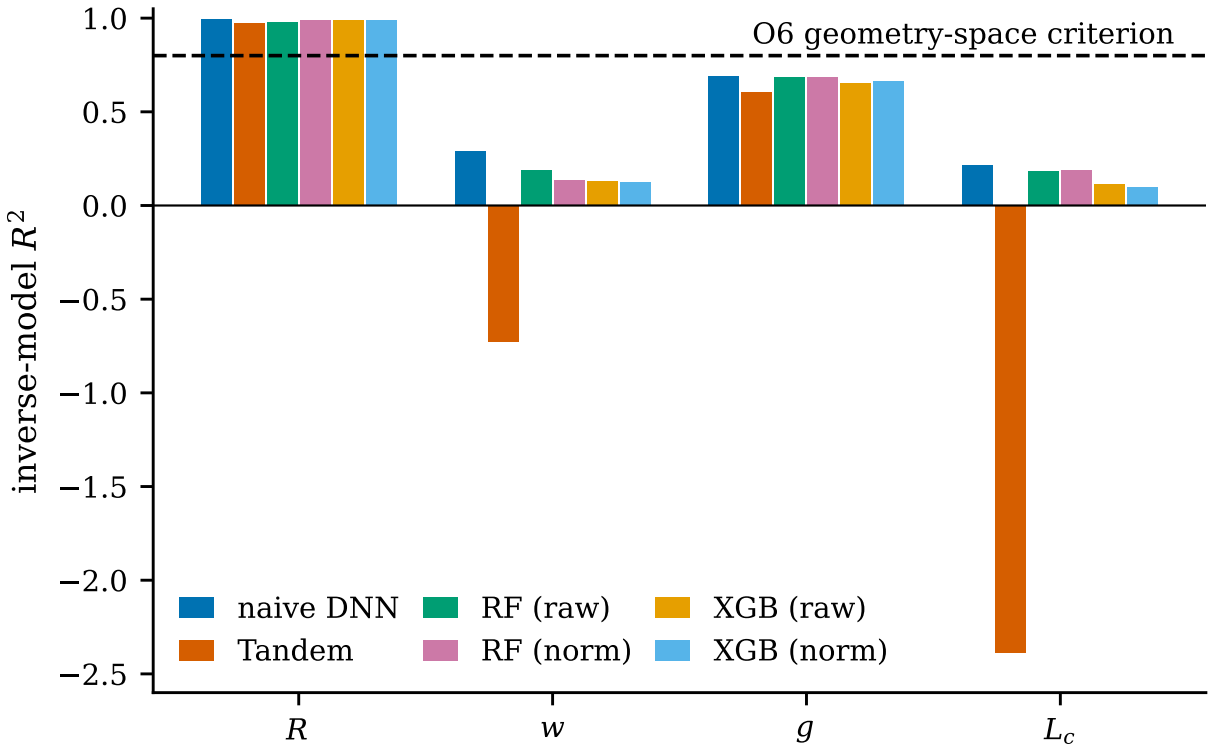


Figure 1: The geometry-space half of the one-to-many test. Only the ring radius is recovered; width, gap and coupling length are not, for every method. The dashed line is the $R^2 \geq 0.80$ criterion. The tandem network's negative values for w and L_c indicate predictions worse than the test-set mean.

therefore impose two effective constraints — one on L, one on θ — on four unknowns, leaving a two-dimensional family of geometries with indistinguishable responses.

That prediction is quantitatively borne out. L is dominated by $2\pi R$, so R is recoverable, and it is ($R^2 = 0.9930$). The remaining three parameters enter only through θ, and none is individually recoverable ($R^2 = 0.13$–0.69 across all methods). Direct confirmation: four geometries with gaps spanning 210 nm to 270 nm and coupling lengths spanning 0.209 µm to 2.353 µm share a loaded Q of 14 659.8 to the last digit. The naive network recovers θ itself to 1.824%, with 96.5% of its predictions within 5% of the correct level set. It is not failing; it is landing on the right level set and being scored on its position along it.

A systematic search for degenerate pairs. The four-geometry example above was found by inspection, so we also searched for such cases systematically. Among all pairs of the 1000 primary-design rows we kept those whose targets are practically indistinguishable: FSR within 1%, Q_L within 5%, IL within 5% and λ_{res} within 0.5 nm. These tolerances are our choice, not derived from a measurement uncertainty. Nineteen pairs qualify. Measured with the RMS-RGE convention, the geometric difference between the two members has a median of 19.4%, a 90th percentile of 31.4% and a maximum of 45.8%; it exceeds 15%, the aggregate error of the naive network, in 13 of the 19 pairs and exceeds 30% in three. The most different pair has coupling lengths of 2.63 and 0.18 µm and gaps of 293 and 207 nm, yet FSR of 6.786 and 6.844 nm, Q_L of 11 185 and 11 550, and IL of 0.292 and 0.291 dB. A model that sees only the targets must return the same geometry for both members of a pair, so by the triangle inequality at least one of them is missed by roughly half

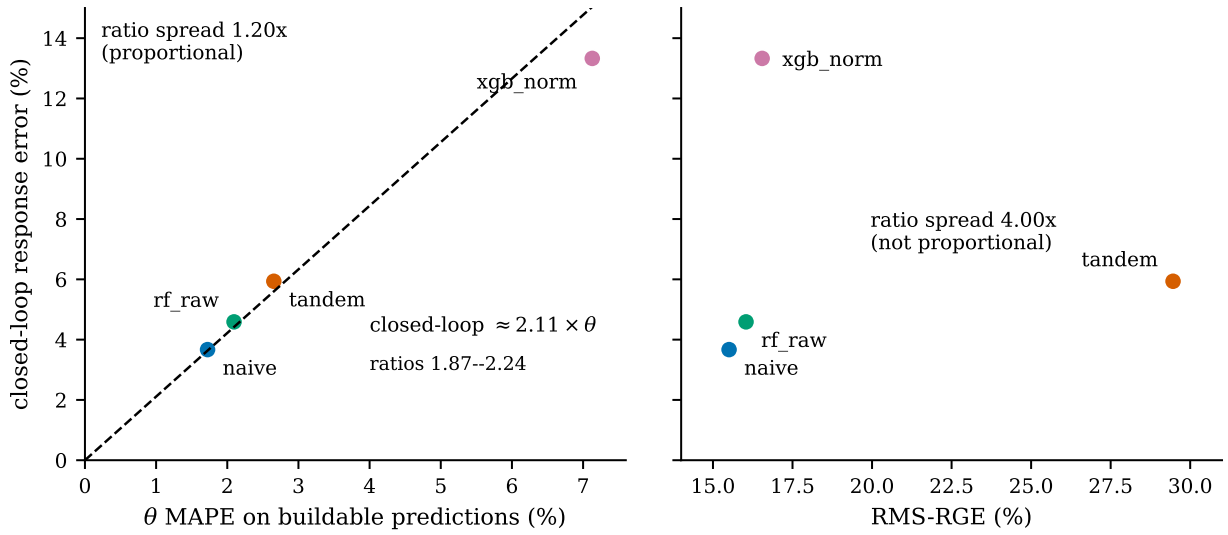


Figure 2: Closed-loop response error against two candidate metrics, for the four models whose predictions were re-simulated. Left: error in the coupling phase θ, evaluated on buildable predictions so that the two axes cover the same designs. Right: RMS-RGE. The ratio of closed-loop error to θ error is constant to within a factor 1.20; formed with RMS-RGE it varies by a factor 4.00.

the pair difference (approximately, because each relative error is normalised by its own true value). For the pair above that is about 23%, more than the naive network's total error. Pairs are rare because 1000 points are sparse in a four-dimensional space, so the count shows that degeneracy exists and says nothing about how common it is. Because this verdict rests on labels produced by the coupled-mode assembly, and Section 2.4 measured that assembly's group index to sit about 3% below full 3-D FDTD, it is fair to ask whether a label error of that size could manufacture or conceal the degeneracy. We tested it directly. The inverse model is trained on FSR labels rescaled by ±3% and then asked for geometries matching *unshifted* targets — the mismatch between the labels a model learns from and the response a user requests is what a biased label actually costs. (Shifting both sides instead is invariant: tree splits rescale with the feature and standardisation absorbs a uniform factor, so no model in the pipeline can respond to it.) Across ±3% the aggregate geometric error moves by 0.65 percentage points and the reconstructed response error by 0.13, and the count of parameters reaching $R^2 \geq 0.80$ never leaves one of four: R stays at 0.958–0.961 while w, g and L_c stay at 0.16–0.68. The degeneracy is structural, as the constraint-counting argument requires — a uniform shift in one label cannot change how many independent constraints four targets impose. This test uses a plain random-forest inverse as its vehicle, so its absolute error level is not comparable with Table 4; only the differences across the three label scalings are meaningful. It also probes the simplest form the label error could take, a uniform scaling, whereas the measured discrepancy is geometry-dependent with a mild wavelength trend.

3.3 Geometry-space error does not predict response-space error

Section 3.2 showed that geometric error and response error disagree about whether a model works. Table 7 shows that they also disagree about which models work, and by how much.

The naive network and XGBoost (normalised) differ by 1.04 percentage points in RMS-RGE — 15.506% against 16.546%, a difference a reader would reasonably call negligible. In closed loop they differ by a factor of 3.6 (3.669% against 13.329%; bootstrap 95% interval for the ratio 2.3–6.3 over

Table 7: Closed-loop re-verification, four models, twenty targets each. All clear the 20% response-error bar. θ error is evaluated on buildable predictions, matching the designs over which closed-loop error is measured. The λ_{res} column is a percentage of about 1550 nm; because the targets lie within $\pm 0.5\%$ of 1550 nm it is not informative on its own (see Section 3.7 for the value in nm and a no-skill baseline).

Method	RMS-RGE (%)	θ err. (%)	closed-loop error, per target (%)				RMS (%)
			λ_{res}	FSR	Q_L	IL	
naive DNN	15.506	1.723	0.203	0.980	4.037	3.594	**3.669**
RF (raw)	16.039	2.095	0.208	2.994	4.058	5.011	4.589
Tandem[†]	29.461	2.654	0.190	0.745	4.980	7.110	5.936
XGB (norm)	16.546	7.129	0.165	0.970	14.684	15.382	13.329

[†]Over the 8 of 20 predictions that were buildable; see Section 3.4.

the twenty targets). Had only RMS-RGE been measured, these two models would have been called equivalent, and the conclusion would have been wrong by nearly a factor of four on the quantity that determines whether a design works. Closed-loop error is computed from twenty test targets spaced evenly through the test set; bootstrap 95% intervals on the RMS are 2.4–5.1% (naive DNN), 3.4–5.7% (RF raw), 9.1–17.1% (XGBoost norm) and 2.3–8.7% (tandem, 8 targets).

The degeneracy suggests why, and θ makes it measurable. An error that moves along a θ level set yields a different valid design; one that moves across it yields a wrong response. Figure 2 tests this on the four models whose predictions were re-simulated. Closed-loop error is proportional to θ error, with the four ratios spanning 1.87–2.24, a factor of 1.20. The same ratios formed with RMS-RGE are 0.20–0.81, a factor of 4.00. Part of this is expected: because $\kappa^2 = \sin^2\theta$, a small relative error in θ produces roughly twice that relative error in κ^2, and hence in Q_L and IL. The measured ratios (about 2.1) are consistent with that. The result rests on four models and is a consistency check on the constraint-counting argument, not an independent statistical test.

Two features of that comparison matter for its strength. The first is that model family is controlled: Random Forest (raw) is an ensemble method with θ error 2.095% and lands at 4.589%, beside the network's 3.669% and far from XGBoost's 13.329%. Within the tree family alone, θ error varies by a factor 3.4 and closed-loop error by 2.9, while RMS-RGE differs by 3% — so there is no architecture explanation available. The second is that ordering alone would not have discriminated: RMS-RGE happens to rank three of these four models correctly. What it cannot do is say how much worse one is than another, which is what a design metric is for.

3.4 Physical realisability

Neither RMS-RGE nor θ error detects a prediction that cannot be fabricated, and the differences between architectures here are large.

The tandem network produced unbuildable geometries for twelve of the twenty closed-loop targets: eleven with negative coupling length, down to -0.881 µm, and one with a width of 397.8 nm. Over the full test set its buildable fraction is 33.5% at the reported initialisation, and 11.6–83.8% over five initialisations (Section 3.5). The mechanism is visible in its loss: the tandem network is penalised only through the frozen forward surrogate, which will evaluate $L_c = -0.88$ µm and return a plausible response. Nothing in the objective encodes that the design space has edges. The naive network, which regresses against real geometries and is therefore pulled toward a region where all targets are valid, leaves the box on 4.6% of the test set.

Both tree families produced buildable geometries in every case. For Random Forest this is guaranteed rather than fortunate: its prediction is a mean of leaf values and hence a convex

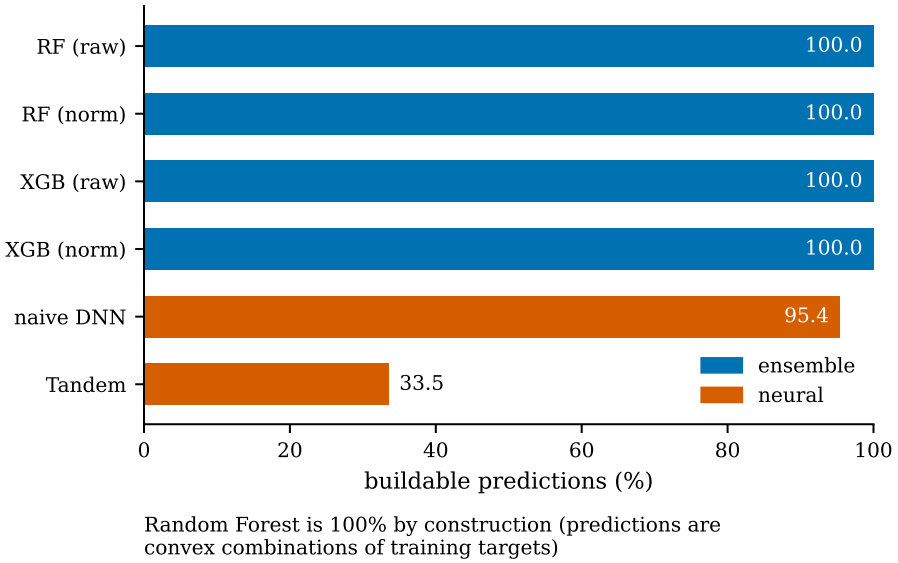


Figure 3: Fraction of test-set predictions lying inside the design box. The distinction is architectural, and invisible to both metrics of Section 3.3.

combination of training targets, so it lies in their convex hull, and the design box is convex. Gradient boosting is additive rather than convex and carries no such guarantee, but violated the box nowhere here.

The practical consequence is that θ error and realisable fraction are complementary and both are needed; neither subsumes the other. The tandem network illustrates why. On its buildable predictions its closed-loop error is 5.936%, better than XGBoost's, and its θ error is a respectable 2.654%. Judged on those numbers alone it looks like a competitive model. Sixty per cent of its designs cannot be made.

3.5 Method comparison and sample efficiency

Because a curve read at a single random seed cannot distinguish an ordering from initialisation noise, the whole sweep was repeated over five seeds (42–46) and is reported as mean ± standard deviation (divisor n; with divisor $n-1$ every spread in this section would be 11.8% larger). Two methods are taken to be separated only when the gap between them exceeds the pooled seed spread.

In the forward direction at the full 805-sample pool the network is most accurate, at $3.776 \pm 0.282\%$ against XGBoost's $5.115 \pm 0.115\%$: a gap of 1.34 against a pooled spread of 0.30, so the ordering is real. Below 300 samples it is *not*. XGBoost averages 12.815% and the network 14.303%, a gap of 1.49 against a pooled spread of 3.09, driven by the network's seed-to-seed variability at small n (± 3.07 at $n = 100$). The two are comparable in that regime and we do not claim a winner. A single-seed reading of the same sweep shows XGBoost ahead by 4.4 points and would support the opposite, stronger statement; it does not survive repetition.

In the inverse direction the naive network is lowest at the smallest and the largest training sizes (16.475% against the random forest's 16.875% below 300 samples; $15.500 \pm 0.069\%$ against $15.853 \pm 0.016\%$ at full size), separated over seeds by the pooled-spread rule. Two cautions apply. The random forest's own curve is not monotone (15.473% at $n = 600$, 15.853% at $n = 805$, a change of 0.38 percentage points), which is as large as the margins above, so the ordering is established across initialisations but not across draws of the data. And test-set sampling uncertainty is larger

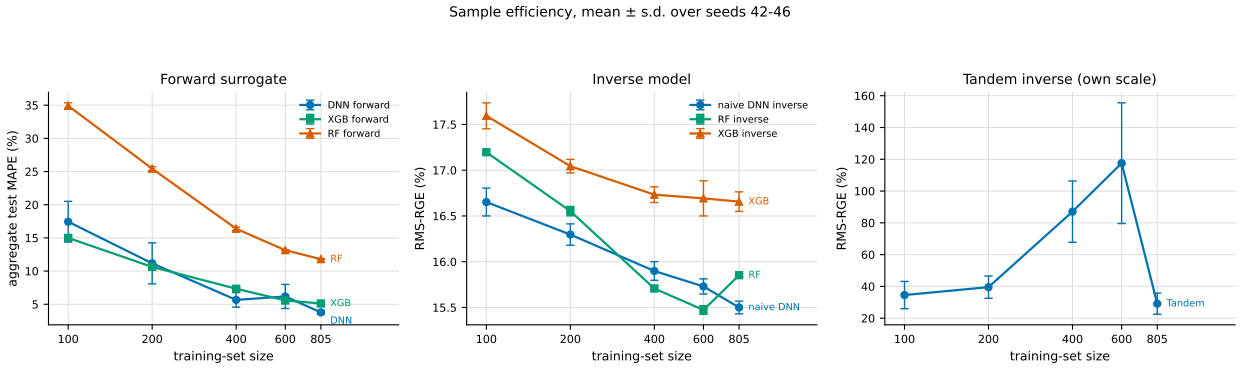


Figure 4: Sample efficiency, mean ± standard deviation over five seeds (42–46). Left: forward direction — the apparent XGBoost advantage below 300 samples is inside the seed spread and is not claimed, while the network's advantage at the full pool is outside it. Centre: inverse direction, where the naive network's margin over the random forest is reproducible across seeds, though not separated over test rows at full size (text). Right: the tandem network on its own scale, spanning 29–118% and non-monotonic in training size; plotted on the centre panel's axis it would flatten the differences that panel exists to show.

than seed spread: 95% bootstrap intervals over the 173 test rows are about ±1.0 to 1.2 percentage points for every model (naive DNN 15.51%, interval 14.47–16.52%; RF (raw) 16.04%, 14.96–17.12%). Paired over the same resampled rows, the naive network's advantage at full size is −0.53 points against RF (raw) (interval −1.23 to +0.14), −0.33 against RF (norm) (−1.12 to +0.45), −0.95 against XGBoost (raw) (−1.94 to +0.06) and −1.04 against XGBoost (norm) (−2.04 to −0.03). Only the last interval excludes zero, and only just, so at full size the naive network is not shown to be more accurate than the random forest on this test set; it is clearly more accurate than the tandem network (−13.96 points, interval −15.72 to −12.04). The O3 shortfall (15.5% against 15%) lies inside the naive network's own interval (bootstrap probability that its RMS-RGE is at most 15%: 0.17), so we report it as "not met at the point estimate", not as missed by a fixed number of standard deviations.

The tandem network's inverse curve is not a learning curve at all: 34.6 ± 8.6, 39.5 ± 7.0, 87.1 ± 19.3, 117.5 ± 38.0 and 29.1 ± 6.7 per cent across the five sizes — non-monotonic in training size, with a standard deviation approaching a third of the mean. This is consistent with the box-escape mechanism of Section 3.4; with nothing anchoring predictions inside the design space, where the network wanders is largely a function of initialisation and data. The tandem architecture is the standard remedy for one-to-many ambiguity in this literature, and, as implemented here — unbounded outputs, and a loss in which λ_{res} contributes 99.3% of the response-space gradient — it is worse than the naive network and its error is erratic across seeds. Whether this reflects the architecture or this implementation of it was tested directly. Four tandem variants were trained over the same five seeds (42–46) through the same frozen forward model (Table 8): the baseline above; a variant with a sigmoid output, so that every prediction lies inside the design box; a variant in which each target's squared error is divided by the forward model's own validation error on that target, which has no free parameter and lowers the weight of λ_{res} to 0.003 of the mean; and both together.

Bounding the output removes both faults of the baseline: every design is buildable, and the seed-to-seed spread of RMS-RGE falls from 5.79 to 0.29 percentage points. It does not close the gap in geometry space: the bounded tandem network stays 3.96 points above the naive network, against a pooled seed spread of 0.21. Weighting alone leaves the geometry unchanged (31.35 ± 7.74%, 30.6% buildable) but improves the reconstructed response, from 4.34% to 3.31%, mostly through Q_L. With both changes, the coupling-phase error (1.92 ± 0.11%) and the reconstructed-response error (3.61%)

Table 8: Tandem variants, mean ± standard deviation (divisor n) over five seeds. Response MAPE is the mean over the four targets of the error between the requested response and the frozen forward model's response to the predicted geometry. The naive row gives RMS-RGE over five seeds; its other columns are from the headline run (seed 42) over the same 173 test rows.

Variant	RMS-RGE (%)	θ MAPE (%)	buildable (%)	response MAPE (%)
naive DNN	15.500 ± 0.069	1.82	95.4	3.81
tandem, baseline	30.03 ± 5.79	7.36 ± 3.47	37.3 (11.6–83.8)	4.34 ± 0.07
tandem, bounded	19.46 ± 0.29	2.62 ± 0.06	100	4.53 ± 0.04
tandem, weighted	31.35 ± 7.74	8.40 ± 4.95	30.6	3.31 ± 0.04
tandem, bounded + weighted	19.76 ± 0.31	1.92 ± 0.11	100	3.61 ± 0.02

are comparable to the naive network's (1.82% and 3.81%, single run), while RMS-RGE remains about four points higher. A bounded, weighted tandem network is therefore not worse than the naive network in response space or in θ; it is worse only in geometry space, which is the distinction this paper is about. These runs are not the sweep of Figure 4: that sweep retrains the forward network for every seed and initialises the tandem network at seeds 43–47, so its full-size tandem spread ($29.1 \pm 6.7\%$) is a different experiment, with one run in common (29.461%). These variants were not re-simulated in closed loop, so no closed-loop claim is made for them.

A final observation on metric dependence. Ranking parameter difficulty by mean R^2 gives $L_c > w > g > R$, hardest first; ranking by mean RMS-RGE contribution gives $L_c > g > w > R$. The two orderings disagree on w against g. Under either, the coupling gap is *not* the hardest parameter — the coupling length is, with mean $R^2 = -0.2645$ — which refutes the hypothesis that g would dominate by analogy with [6]'s finding for radius.

3.6 An information bound

Objective O3 required, for at least one method, RMS-RGE $\leq 15\%$ *and* closed-loop error $\leq 20\%$. The closed-loop half is met with a wide margin (3.669%). The geometric half is not: the best aggregate RMS-RGE is 15.506%. **O3 is therefore not met**, by 0.506 percentage points.

Whether that shortfall reflects the models or the problem is a separate question, and answering it requires knowing how much geometric information four scalar targets contain. We measure this directly. A bank of 10^6 geometries is drawn over the design box and labelled with the same analytic responder that produced the dataset, subject to the same quality gate. For each test target, the k bank geometries whose responses are closest are averaged, and the RMS-RGE of that average against the true geometry is computed. Because this is the error of one specific predictor, it is an *upper bound* on the Bayes-optimal error.

The asymmetry of that bound governs what can be concluded. An upper bound measured below a threshold proves the true bound is below it; measured above, it proves nothing. Figure 5 shows the estimate falling from 16.063% at $N = 12\,500$ to 14.626% at $N = 10^6$. Since $14.626\% < 15\%$:

> **At $N = 10^6$ the estimate is 14.626%, below the 15% threshold; we therefore have no evidence that the threshold is unreachable.** We do not claim it is reachable: the neighbourhood size k was chosen on the same 173 test targets, a second run ($k = 5$) gives 14.669%, the k-curve is not smooth (14.67% at $k = 5$, 14.93% at $k = 10$), and no sampling interval has been computed.

Two honest qualifications. The estimate is not converged: the optimal neighbourhood size drifts from $k = 25$ to $k = 7$ across the sweep, and the descent accelerates from 0.096 to 0.530 percentage

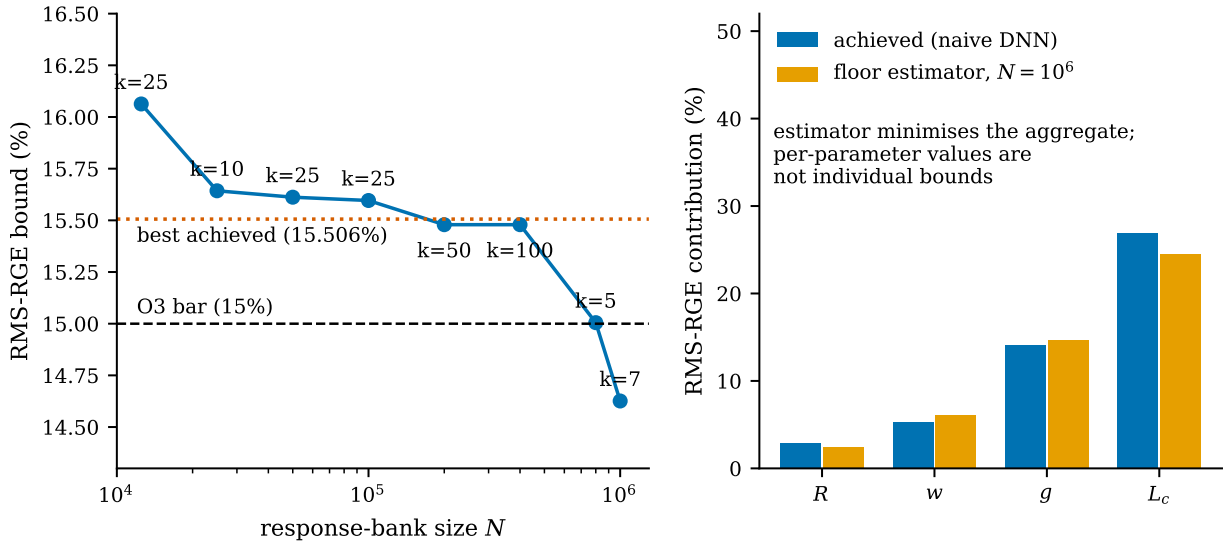


Figure 5: Left: the information bound against response-bank size, with the best neighbourhood size k annotated. The estimate is an upper bound on the true bound, and falls as the bank grows. At $N = 10^6$ it lies below the 15% objective threshold. Right: per-parameter error of the best model against the bound estimator's per-parameter error; these are not individual bounds, since the estimator minimises the aggregate. The headline estimate is from the convergence run ($k = 7$); the per-parameter values on the right are from a separate run at $k = 5$ (14.669%).

points per e-fold of bank size. A shifting optimum means the estimator's own finite-sample bias is still unwinding, so no rate from this sweep describes the true bound, and a three-parameter asymptotic fit to these points is not identifiable. And the per-parameter values in Figure 5 are not individual bounds: the estimator minimises the aggregate and trades parameters against one another, which is why the best model legitimately beats two of them.

We also record a negative decision. Enlarging the training set with additional rows drawn from the *fitted* analytic model lowers measured RMS-RGE below 15%. Those rows add sampling density, not physics — the dataset would still carry the information of 200 coupler simulations — and the closed-loop error against FDTD would not improve. We therefore do not report O3 as met by that route, and record here that it was available and declined.

3.7 Translation to co-packaged-optics scenarios

Objective O7 asked which methods meet a CPO scenario's specification within its stated tolerance. Table 9 answers: **none, in either scenario, within the uncertainty of the ring model**. Each candidate's coupler was re-simulated in 3-D FDTD; the ring response was then assembled analytically, so the verdicts inherit the biases of Section 2.4.

Eleven of the twelve failures involve resonance placement. The single exception is Random Forest (raw) in Scenario A, which placed the resonance at 1548.058 nm, inside the ±2.0 nm window, and then missed FSR by 0.13 nm. FSR, Q_L and IL are otherwise largely satisfied; only XGBoost undershoots Q_L in Scenario B.

This failure is physical, not algorithmic, and the numbers say so without interpretation. The naive network's closed-loop λ_{res} error is 3.15 nm MAPE, 4.32 nm RMS and 10.9 nm at worst. For reference, always predicting the dataset's mean resonance wavelength gives 2.66 nm RMS on the same twenty targets, and a randomly chosen geometry about 4.4 nm; the models (3.35 nm to 4.33 nm

Table 9: Re-simulated scenario verdicts. Each candidate geometry was re-simulated with a fresh 3-D FDTD coupler run. Tolerance on λ_{res} is half the channel spacing: ± 2.0 nm in A, ± 0.4 nm in B. If the ring FDTD is taken as the reference, the analytic FSR is about 2.4% low: RF (raw) in Scenario A would then pass FSR (about 16.3 nm), and its λ_{res} of 1548.06 nm lies inside the ± 2 nm window by 0.06 nm, less than the 0.8 nm to 1.1 nm λ_{res} error of the analytic model, so its verdict is undecided. Q_L above about 4300 (all Scenario B candidates) is not validated.

Method	λ_{res} (nm)	FSR (nm)	Q_L	IL (dB)	Verdict
Scenario A — 4 channels, 4 nm *grid; FSR* ≥ 16 nm, $Q_L \in [1500, 8000]$, *IL* ≤ 1.5 dB					
naive DNN	1546.892	16.351	3995	0.352	fail: λ_{res} −3.11 nm
Tandem	1543.385	16.640	3826	0.345	fail: λ_{res} −6.61 nm
RF (raw)	1548.058	15.869	5425	0.462	fail: FSR −0.13 nm
RF (norm)	1543.823	15.901	5917	0.508	fail: λ_{res} −6.18, FSR −0.10
XGB (raw)	1557.673	16.298	3708	0.322	fail: λ_{res} +7.67 nm
XGB (norm)	1547.746	16.111	3890	0.336	fail: λ_{res} −2.25 nm
Scenario B — 8 channels, 0.8 nm *ITU grid; FSR* ≥ 6.4 nm, $Q_L \geq 7750$, *IL* ≤ 1.0 dB					
naive DNN	1548.663	6.458	8575	0.207	fail: λ_{res} −1.34 nm
Tandem	1548.770	6.344	8196	0.190	fail: λ_{res} −1.23, FSR −0.06
RF (raw)	1549.537	8.446	10571	0.361	fail: λ_{res} −0.46 nm
RF (norm)	1548.733	6.474	12291	0.301	fail: λ_{res} −1.27 nm
XGB (raw)	1550.934	6.405	6070	0.147	fail: λ_{res} +0.93, Q_L −1680
XGB (norm)	1552.793	6.424	5816	0.141	fail: λ_{res} +2.79, Q_L −1934

RMS) therefore place the resonance no better than predicting the mean. Scenario A allows 2.0 nm and Scenario B 0.4 nm, so the best model misses by factors of two and eleven respectively. Working back through $\mathrm{d}\lambda_{\mathrm{res}}/\mathrm{d}R \approx 0.15$ nm/nm, hitting Scenario A would require the radius controlled to about 13 nm and Scenario B to about 3 nm — below realistic lithographic tolerance, let alone inverse-model precision.

Three views of one effect agree: the sensitivity analysis of Section 3.1, the closed-loop re-simulation of Section 3.3, and the scenario translation here. All reflect that the resonance is the nearest one to 1550 nm, so it wraps every FSR. Inverse design delivers the *shape* of the response — free spectral range, loaded Q, insertion loss — and cannot deliver absolute resonance position. A CPO deployment must place channels by post-fabrication thermal or carrier trimming.

3.8 What the trimming costs

That conclusion is only useful if the trimming it demands is practical, so we quantify the demand. The thermo-optic sensitivity is solved for rather than assumed: each material index is perturbed in turn and the mode re-solved, giving $\Gamma_i = \partial n_{\mathrm{eff}}/\partial n_i$ directly. Perturbing both materials together reproduces the sum of the separate perturbations to within 0.17% across all six widths, so the response is linear and the two sensitivities combine. With the literature $\mathrm{d}n/\mathrm{d}T$ of silicon and oxide — the same class of input as the Palik index data used throughout — and n_g from the mode sweep,

$$\frac{\mathrm{d}\lambda_{\mathrm{res}}}{\mathrm{d}T} = \frac{\lambda}{n_g}\left[\Gamma_{\mathrm{Si}}\frac{\mathrm{d}n_{\mathrm{Si}}}{\mathrm{d}T} + \Gamma_{\mathrm{SiO_2}}\frac{\mathrm{d}n_{\mathrm{SiO_2}}}{\mathrm{d}T}\right], \tag{3}$$

which evaluates to 0.0700 nm/K to 0.0734 nm/K over 400 nm to 500 nm of width. Published silicon microring tuning is 0.07 nm/K to 0.08 nm/K; nothing here was fitted to that, so the agreement is an independent check.

Two physical constraints govern the budget. A thermo-optic heater shifts a resonance to the red only, and the resonance comb repeats every FSR, so a ring is tuned onto whichever resonance is nearest above it. A ring designed with a deliberate blue bias δ therefore needs a shift of $(\delta - e_i) \bmod \mathrm{FSR}_i$, where e_i is its closed-loop λ_{res} error. Choosing δ to minimise the worst case is a covering problem on a circle of circumference FSR, solved here by direct search. The requirement is bounded by one FSR — about 227 K for these geometries — however poorly a model places λ_{res}.

Table 10: Thermal trimming demand over the twenty closed-loop targets. *Biased* assumes the optimal design offset; *unbiased* assumes none. Power uses a thermal resistance of 3 K/mW, a device parameter that scales the power columns linearly and the temperature columns not at all. Circular spread is the smallest arc of one FSR containing all errors; the mean FSR is 9.74 nm.

		biased				unbiased	
Method	RMS λ_{res} (nm)	spread (nm)	circ. spread (nm)	max ΔT (K)	max (mW)	mean ΔT (K)	max ΔT (K)
Naive DNN	4.332	18.940	6.898	125.0	41.7	80.1	227.5
RF (raw)	4.015	16.433	7.873	**121.4**	40.5	70.1	190.4
XGBoost (norm)	3.346	12.371	7.114	131.9	44.0	56.7	204.9
Tandem realises only 8 of 20 predictions and is excluded from the ranking: scoring it on the subset it happened to realise is selection after the fact.							
Tandem ($n = 8$)	3.638	9.082	5.822	76.0	25.3	58.0	104.1

Table 10 gives the result. The best fully-evaluated model needs a heater spanning 121 K, drawing 18.2 mW on average across rings and 40.5 mW at worst — 73 mW static for a four-channel link and 146 mW for eight channels, at typical draw. Under the stated assumptions (one heater per ring, 3 K/mW) trimming is feasible; we do not compare 18 mW per ring with a CPO power budget, so we make no claim that it is affordable.

The errors of every model fill most of one FSR: the circular spread is 71–81% of the 9.74 nm mean FSR, so every model needs a heater range of 121 K to 132 K at the optimal design bias. Demand differs by 8% between the three fully evaluated models, from twenty targets and one trained model each, which does not separate them. The ranking depends on the statistic: the random forest has the smallest maximum biased temperature rise (121.4 K), XGBoost the smallest mean (52.8 K biased, 56.7 K unbiased). We therefore draw no conclusion about which model is cheapest to trim, and note that at the present placement accuracy the demand of every model is close to the range needed to tune across one mean FSR (about 135 K).

4 Discussion

The central result is that a question treated as empirical in the inverse-design literature was in fact definitional. Asking whether a naive inverse network "converges acceptably" has no answer until one says in which space the error is measured, and the two available answers differ here by a factor of four on the same predictions. For a degenerate design space the geometry-space question is not merely harder, it is the wrong question: a model that lands on the correct level set has produced a valid design, and penalising its position along that set measures nothing a user cares about.

That observation generalises beyond microrings. Wherever the forward map compresses several geometric degrees of freedom into fewer physical ones — which is to say, wherever inverse design is interesting — the same split should be expected, and the recommendation follows directly: report closed-loop response error, and report it alongside a physically meaningful invariant of the

degeneracy rather than a raw geometric distance. Our θ is such an invariant for this device; the general construction is to identify the quantity through which the degenerate parameters act and measure error in it.

The realisability result adds a second axis that neither error metric sees. It also puts a specific cost on the tandem architecture. The tandem network was introduced [3] precisely to handle degeneracy, and in response space it does: its closed-loop error on buildable designs, 5.936%, is better than XGBoost's. But measuring the loss only through a surrogate removes the sole pressure keeping predictions inside the manufacturable domain, and two thirds of its designs leave it in this implementation. A tandem network for a bounded design space needs an explicit constraint, and reporting realisable fraction would have made this visible in earlier work. A sigmoid-bounded variant confirms it: 100% of its designs are buildable over five seeds, and its RMS-RGE spread falls from 5.79 to 0.29 points (Section 3.5).

The information bound changes how a missed threshold should be read. RMS-RGE targets in this literature are quoted without reference to how much geometric information the chosen targets contain — [12] reports 3.46% and 5.14% under the $\{R, w, g\}$ convention, against 8.878% here under the same convention, in a design space one to two orders of magnitude different in three of four dimensions. Such numbers are not commensurable, and a bound computed per design space is what makes them interpretable. Here the bound establishes that the shortfall is addressable; in another design space the same computation might show a target to be unreachable, which is equally useful to know before pursuing it.

5 Limitations

1. **The ring validation is not mesh-converged, and cannot be on the available hardware.** Simulated Q_L depends on mesh spacing (Section 2.4). A third, finer mesh for these geometries exceeds the 15.7 GB of system memory available, so the Richardson extrapolation rests on two mesh sizes and assumes second-order convergence, which two points cannot verify.

2. **Ring validation is at small radius only.** All nine runs are at $R = 5.20\,\mu\text{m}$. A 10 µm ring requires about 20 GB of solver memory at this mesh and a 14 µm ring about 36 GB, so the radius range could not be spanned. Small radius is where bend loss is largest and the loss model most stressed, but the validation does not establish accuracy at large radius.

3. Q_L **accuracy is established only below** $Q_L \approx 4300$, to within 2%. Above it the simulated Q_L is 17–27% below the analytic value at the 32 nm and 40 nm meshes (the analytic value is 21–38% above the simulation) and the cause is not established. 84.7% of the labels lie above $Q_L = 4300$. **IL is not validated at all.**

4. **The external validation covers FSR only.** It tests the mode solver and the length convention. The coupling model, the bend-loss model, Q_L and IL are not externally validated against measurement.

5. **The information bound is not converged.** It is a valid upper bound at every bank size, which is enough for the one-directional conclusion drawn, but its asymptote is not determined by the sweep performed.

6. λ_{res} **is unlearnable within this parameterisation.** An alternative target — for example resonance offset from a grid wavelength, or azimuthal mode order — might be learnable. We measured one reparameterisation candidate and found that within an already matched set the

mean $|\mathrm{corr}(\lambda_{\mathrm{res}}, g)|$ is 0.095, so λ_{res} is close to redundant with FSR for geometry recovery; a systematic study was not performed.

7. **Both couplers share a gap and coupling length.** Asymmetric couplers are a valid extension and would change the degeneracy structure.

8. **Closed-loop verification uses twenty targets per model**, the stated minimum, and four of six models; only the naive network was extended to a hundred targets. The two unmeasured models are ones for which the two candidate metrics make predictions differing by less than one percentage point.

9. **No fabrication.** Every result is simulation-based. The resonance-placement conclusion in particular predicts that trimming is required; it does not demonstrate a trimmed device.

10. **Five seeds is a small sample for a spread.** The separation test of Section 3.5 uses the standard deviation over five seeds, which is itself uncertain; a borderline comparison could be decided either way by a larger repetition. Only the clearly separated and clearly unseparated cases are treated as results.

11. **Label fidelity is tested only against a uniform shift.** The ±3% perturbation of Section 3.2 leaves the O6 verdict unchanged, but a uniform rescaling is the simplest form the measured discrepancy could take; a geometry-dependent label error was not tested and would require regenerating the dataset from full 3-D simulation.

12. **The ring's group index is about 3% below the mode solver's, and the cause is not established.** Round-trip length, bend effects, FSR extraction and the n_g polynomial are each excluded by measurement (Section 2.4). The residual is a quantified, reproducible disagreement between the coupled-mode assembly and full 3-D FDTD, stable across three geometries. Every FSR-derived quantity, including the L_c recovered from the FSR constraint, inherits it.

13. **Bend loss is not resolvable by the eigenmode route.** The imaginary part of n_{eff} returned by the FDE solve at these radii is negative at every grid point and of magnitude 10^{-12} dB/cm to 10^{-5} dB/cm, i.e. at the solver's numerical floor. The wavelength-independent bend-loss model therefore stands uncorrected, and a wavelength-resolved replacement would require 3-D FDTD of a bend section.

14. **No independent expert review of the physical methods** was obtained prior to this draft.

15. **The ring response in the closed-loop and scenario evaluations is analytic;** only the coupler is re-simulated.

16. **Test-set sampling uncertainty is not covered by the five-seed spreads** (173 targets for geometric error, 20 for closed-loop error); bootstrap intervals are given where computed.

17. **The headline tandem baseline has unbounded outputs and an unweighted loss dominated by λ_{res}.** Bounded and weighted variants were evaluated for geometry accuracy, θ error and buildable fraction (Section 3.5) but not in closed loop; the closed-loop figures in Table 7 are for the baseline at a single initialisation.

6 Conclusion

For a four-parameter add-drop silicon microring resonator, a naive inverse neural network fails the geometry-space convergence criterion and passes the response-space one. Both findings reported

in the prior literature are therefore consistent with these results, and the apparent disagreement between them was a difference of metric rather than of result. The reason is a two-dimensional degeneracy that follows from counting constraints: four scalar targets impose two effective conditions, one on round-trip length and one on a single coupling phase.

Three consequences for practice follow. Geometric error does not predict response error — two models within 1.04 percentage points of each other in RMS-RGE differ by a factor of 3.6 in closed loop, while error in the coupling phase predicts closed-loop error to within a factor 1.20 across three architecture families. Physical realisability is a separate and architecture-dependent property that neither metric detects, ranging from a provable 100% for Random Forest to 33.5% for a tandem network at the reported initialisation (and 100% once its outputs are bounded). And an information bound computed per design space is what makes an accuracy target interpretable; here the point estimate gives no evidence that the missed 15% threshold is fundamental.

For co-packaged optics specifically, the useful conclusion is a negative one seen three ways: inverse design can deliver free spectral range, loaded Q and insertion loss, and cannot place an absolute resonance wavelength to channel-grid accuracy. Channel assignment must be achieved by post-fabrication trimming, and an inverse-design flow for CPO filters should be specified on that basis.

Data and code availability

The complete simulation, training and evaluation pipeline is a single Python script, and every reported number is produced by one of its documented modes. The dataset, the fitted models, the response bank and all report files will be released with the manuscript.

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